

|           |                                                                                                                                          |           |
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|           |                                                                                                                                          |           |
| 1(a)      | force acting between two masses<br><b>or</b><br>force on mass due to another mass<br><b>or</b><br>force on mass in a gravitational field | <b>B1</b> |
| 1(b)      | arc length = $r\theta$<br><br>$d = 1.5 \times 10^{17} \times 1.2 \times 10^{-5} = 1.8 \times 10^{12} \text{ m}$                          | <b>A1</b> |
| 1(c)(i)   | $\omega = 2\pi / T$                                                                                                                      | <b>C1</b> |
|           | $= 2\pi / (44.2 \times 365 \times 24 \times 3600)$                                                                                       | <b>A1</b> |
|           | $= 4.5 \times 10^{-9} \text{ rad s}^{-1}$                                                                                                |           |
| 1(c)(ii)  | gravitational forces are equal<br><b>or</b><br>centripetal force about P is the same                                                     | <b>C1</b> |
|           | $M_1 x \omega^2 = M_2 (d - x) \omega^2$ so $M_1 / M_2 = (d - x) / x$                                                                     | <b>A1</b> |
| 1(c)(iii) | $x = 0.4d$                                                                                                                               | <b>C1</b> |
|           | $6.67 \times 10^{-11} \times M_1 = (1.0 - 0.4) \times (1.8 \times 10^{12})^3 \times (4.5 \times 10^{-9})^2$                              | <b>C1</b> |
|           | $M_1 = 1.1 \times 10^{30} \text{ kg}$                                                                                                    | <b>A1</b> |